Lecture 16 (*March 29*)

Reading: Section 8.1
Viewing: Lecture 30

We will now introduce the ultimate level of abstraction in our course, which will finally explain the meaning of matrix multiplication. Recall n–dimensional space, which as a set is given by:

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \text{ where } x_1, \dots, x_n \in \mathbb{R} \right\}$$

Definition 16. *A **linear transformation** is a function:*

$$\phi : \mathbb{R}^n \to \mathbb{R}^m \tag{129}$$

which respects the vector space structures of $\mathbb{R}^n$ and $\mathbb{R}^m$, namely addition and scalar multiplication:

$$\phi(\boldsymbol{v} + \boldsymbol{v}') = \phi(\boldsymbol{v}) + \phi(\boldsymbol{v}') \qquad \textit{and} \qquad \phi(\alpha \cdot \boldsymbol{v}) = \alpha \cdot \phi(\boldsymbol{v}) \tag{130}$$

for any vectors $\boldsymbol{v}, \boldsymbol{v}' \in \mathbb{R}^n$ and any scalar $\alpha \in \mathbb{R}$.

Let us consider a linear transformation ϕ as in (129), and express it in terms of the basis:

$$\boldsymbol{e}_j = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

(whose only entry 1 is on the j–th row). The value $\phi(\boldsymbol{e}_j)$ will be a vector in $\mathbb{R}^m$, and therefore a linear combination of the basis vectors $\boldsymbol{e}_i$. This means that there exist constants $a_{ij} \in \mathbb{R}$ such that:

$$\phi(\boldsymbol{e}_j) = \sum_{i=1}^{m} a_{ij} \boldsymbol{e}_i \qquad \forall j \in \{1, \dots, n\} \tag{131}$$

But once you know the numbers a_{ij}, the linear transformation ϕ is uniquely determined! Indeed, since any vector $\boldsymbol{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$ is a linear combination of the $\boldsymbol{e}_i$'s, formulas (130) and (131) imply:

$$\phi(\boldsymbol{v}) = \phi(v_1 \boldsymbol{e}_1 + \dots + v_n \boldsymbol{e}_n) = \sum_{j=1}^{n} v_j \phi(\boldsymbol{e}_j) = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} v_j \boldsymbol{e}_i$$

which completely determines $\phi(\boldsymbol{v})$ for any $\boldsymbol{v}$. In column matrix form, this relation reads:

$$\phi \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} a_{11} v_1 + \dots + a_{1n} v_n \\ \vdots \\ a_{m1} v_1 + \dots + a_{mn} v_n \end{bmatrix} \tag{132}$$

Does this look familiar? Indeed, ϕ acts on column vectors precisely like the matrix:

$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \tag{133}$$

multiplies vectors. We conclude the following:

Fact 11. *There is a one-to-one correspondence between linear transformations and matrices, in that the matrix* (133) *represents the linear transformation* (132)*. In symbols, this means:*

$$\boxed{\phi(\boldsymbol{v}) = A\boldsymbol{v}} \tag{134}$$

for all vectors $\boldsymbol{v}$*.*

Let us discuss the 2×2 case in detail, by showing which geometrically relevant linear transformations $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ correspond to which matrices:

- $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is reflection in the line $\{x = y\}$
- $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ is projection onto the x-axis
- $\begin{bmatrix} \lambda & 0 \\ 0 & \mu \end{bmatrix}$ is a scaling by a factor of λ in the x direction and by a factor of μ in the y direction
- $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is counter-clockwise rotation by the angle θ around the origin
- $\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}$ is a shearing parallel to the x axis

As you can see from these examples, linear transformations send lines to lines and geometric shapes of a certain kind (triangles, quadrilaterals, ellipses) to shapes of the same kind. Another big class of linear transformations are projections, and formula (102) basically states that the function:

$$\mathrm{proj}_V : \mathbb{R}^n \to \mathbb{R}^n$$

which projects onto a given m–dimensional subspace V corresponds to the matrix:

$$A(A^T A)^{-1} A^T$$

where the columns of the $n \times m$ matrix A are given by any basis of V.

Note that composition of linear transformations corresponds to multiplication of matrices. In symbols, if the linear transformations ϕ_1, ϕ_2 are given by matrices A_1, A_2, respectively, then:

$$\boxed{\text{the transformation } \phi_1 \circ \phi_2 \text{ corresponds to the matrix } A_1 A_2} \tag{135}$$

For example, rotation by $\frac{\pi}{2}$ degrees followed by a horizontal shearing is given by the matrix:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

Take now $m = n$. If we have a function $\phi : \mathbb{R}^n \to \mathbb{R}^n$, we can talk about its **inverse** $\phi^{-1} : \mathbb{R}^n \to \mathbb{R}^n$: the latter is connected to the former by the condition that if $\phi(\boldsymbol{v}) = \boldsymbol{w}$, then $\phi^{-1}(\boldsymbol{w}) = \boldsymbol{v}$. It may not surprise you that if the linear transformation ϕ corresponds to a square matrix A, then:

$$\boxed{\text{the transformation } \phi^{-1} \text{ corresponds to the matrix } A^{-1}} \tag{136}$$

The function ϕ has an inverse precisely if and only if the matrix A is invertible.

The nullspace and column space of a matrix A also have an interpretation in terms of the corresponding linear transformation $\phi : \mathbb{R}^n \to \mathbb{R}^m$:

$$N(A) = \text{Ker } \phi \qquad \text{and} \qquad C(A) = \text{Im } \phi \tag{137}$$

where the notions of **kernel** and **image** of a linear transformation ϕ are defined by:

$$\text{Ker } \phi = \Big\{ \boldsymbol{v} \in \mathbb{R}^n \text{ such that } \phi(\boldsymbol{v}) = 0 \Big\}$$
$$\text{Im } \phi = \Big\{ \boldsymbol{w} \in \mathbb{R}^m \text{ such that } \boldsymbol{w} = \phi(\boldsymbol{v}) \text{ for some } \boldsymbol{v} \in \mathbb{R}^n \Big\}$$